Project 7/8

Camera settings research

Merel van der Leeden (1103194)

Sandra Hekkelman &
Wouter Volders

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Summary

• Korte beschrijving van de onderzoeksvragen, methode en resultaten.

• Geef heel kort de belangrijkste conclusie(s) van je rapport aan.

Prologue

One of the tasks in the self-driving challenge is to detect white lines on the track and drive between them. Our solution to this problem is to convert the images from the two outer cameras to black and white and apply Canny edge detection. This ensures clear white lines around all of the objects in the frame. In this research paper, experiments will be conducted to investigate how camera resolution and frame rate affect the quality of line detection.

The main questions for this research paper are:

1. How does camera resolution affect line detection?
2. How does camera frame rate affect line detection?

Theoretical framework

How does camera resolution affect line detection?

What is camera resolution?

Camera resolution refers the number of pixels in a camera frame. Pixels are tiny dots that form a digital image. The number of pixels determins how detailed and clear the image appears. The more pixels in an image the clearer the actual image is.

For example:

This is an image of an apple with the resolution of 225 x 225 pixels.	
This is the same image of the apple but than with the resolution of 90 x 90 pixels.	

The following attachments were used in writing this text:

Attachment 1.

What effect can resolution have on a video?

As stated previously, a higher resolution is equal to more pixels. If you have a higher resolution in a video, than the video would be sharper and more detailed.

How does camera frame rate affect line detection?

What is camera frame rate?

Camera frame rate refers to the number of frames captured per second. The frame rates 30 fps (frames per second) and 60 fps are discussed in this research paper. For instance, a frame rate of 30 fps means that every second the camera take 30 pictures and plays them after each other to create a second of film. Those pictures are called frames. In the same way 60 fps means that the camera take 60 frames very second.

What effect can frame rate have on a video?

The more frames you have every second the smoother the image is. Within the scope of this project the camera's are being used for line detection and object detection. Both of these topics require a camera that can capture enough frames to ensure a smooth motion and reduce blur. Bur is created by a low frame rate.

Method

Type of research: experimental.

The results of the following experiments will help answer the main research questions.

Experiment resolution

Question: How does camera resolution affect line detection?

Why this experiment:

This experiment is performed to determine what camera resolution give the best results for line detection.

Supplies:

1. Cubi NUC AI+, with the drivers for the controller already installed (see attachment 2 for link to the product)
2. HDMI cable
3. Computer screen with a HDMI port
4. Electricity
5. Keyboard and mouse, NOT bluetooth
6. Test code (see attachment 3)
7. Kart, in the same setting as SDC closed challenge 2026
8. Usb stick
9. X-box controller
10. Usb a -> usb c cable
11. Laptop, with anydesk installed

What is the experiment:

The goal of this experiment is to gather mostly identical video footage from the two outer cameras of the kart while using different camera resolutions. To achieve this, follow these steps:

1. Connect the computer screen with the HDMI cable to the NUC.
2. Connect the keyboard and mouse to the NUC.
3. Supply power to the NUC and the screen and turn both devices on.
4. Make sure the test code is on the USB stick.
5. Plug the USB stick into the NUC.
6. Copy the test code from the USB stick onto the NUC.
7. Take out the USB stick.
8. Unplug the NUC from the screen, mouse and keyboard

9. Connect the NUC to the kart and turn both on.
10. Connect the controller with the usb c -> usb a cable to the NUC.

Hypothesis:
The line detection becomes sharper and clearer as the resolution increases.

procedure:
Met het schrijven van de test code voor experiment resolution, zijn we er achter gekomen dat de camera op sommige resoluties frames gaat skippen. Iedere video die is opgenomen heeft de camera 30 seconden aangestaan. Deze test is gedaan op met een windows laptop.
Volgens code : testcodeWindows.py back end dshow

	1920x1080	1280x720	640x480
30	5 seconds	10 seconds	30 seconds
60	2 seconds	5 seconds	15 seconds

Linux:

Code : testcodeLinux.py

Release cameras

Print for resolution, frame rate

Backend msmf → v4l2

	1920x1080	1280x720	640x480
30	30 seconds	30 seconds	30 seconds
60	15 seconds	15 seconds	15 seconds

Results:

Experiment frame rate
Question: How does camera frame rate affect line detection?

Why this experiment:

Supplies:
1. Cubi NUC AI+, with the drivers for the controller already installed (see attachment 2 for link to the product)
2. HDMI cable
3. Computer screen with a HDMI port
4. Electricity
5. Keyboard and mouse, NOT bluetooth
6. Test code (see attachment 3)
7. Kart, in the same setting as SDC closed challenge 2026
8. Usb stick
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10. Usb a -> usb c cable
11. Laptop, with anydesk installed

What is the experiment:
The goal of this experiment is to gather mostly identical video footage from the two outer cameras of the kart while using different camera frame rates. To achieve this, follow these steps:

1. Connect the computer screen with the HDMI cable to the NUC.
2. Connect the keyboard and mouse to the NUC.
3. Supply power to the NUC and the screen and turn both devices on.
4. Make sure the test code is on the USB stick.
5. Plug the USB stick into the NUC.
6. Copy the test code from the USB stick onto the NUC.
7. Take out the USB stick.

Hypothesis:

Procedure:

Results:

Results

• Wat heb je gedaan, wat kwam eruit en omschrijf dit. Hier geef je de ruwe uitkomsten van je onderzoek. Dit betekent dat je er nog geen waardeoordeel over velt. Dit hoofdstuk is enkel feitelijk en omschrijvend.

• Denk hierbij aan meetgegevens, grafieken, enquêteresultaten, informatie afkomstig uit data sheets of andere literatuur. Grote bestanden worden toegevoegd als bijlage, en naar alle bijlages wordt verwezen in de tekst.

• Als je meerdere opties hebt vergeleken om tot een keuze te komen, geef je hier aan in welke maten de verschillende opties aan de requirements voldoen, liefst in een overzichtelijk tabelletje of schema.

Conclusion

• Begin je conclusie met een korte samenvatting en je onderzoeksvraag.

• Hier interpreteer je de resultaten en kom je tot een oordeel. Per onderdeel een conclusie en uiteraard een totaal conclusie (wat is het antwoord op je onderzoeksvraag?)

• Je conclusies worden geschreven in een adviesvorm.

Attachments

1. https://www.unitxlabs.com/blog/camera-resolution-machine-vision-basics-2/
2. https://azerty.nl/product/black-intel-soc-ultra-7-258v-integrated-graphics-n-a-32gbembedded-1tb-ssd-no-hdd-win-11-home-802-11ax-5-2-aircooling-120w-3y/9516612?utm_source=google&utm_medium=cpc&utm_campaign=&&utm_id=1973814685&utm_term=141722565502_&utm_content=645537961146&gad_source=1&gad_campaignid=1973814685&gclid=Cj0KCQjwve7NBhC-

ARIsALZy9HVNztDJLlEg60o97GQgneIqMRgRPNMUCK2XAroVYepSjrMLTfDzUD4aAjWREALw_wcB

3. TEST CODE !!!!!

Changelog

Date	Version	Changed
19/2	1	Which programming language is best for the kart, design made.
5/3	2	Research paper subject has been changed to camera resolution and frame rate. design has been made and prologue is written. A start has been made with theoretical framework and method.
10/3	2.1	Language rewrite
19/3	2.2	Experiments for camera resolution
20/3	2.3	Answered questions about frame rate